

Probability

Addition Rule for Mutually Exclusive Events

$$P(A_1 \text{ or } A_2 \text{ or } \dots \text{ or } A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$$

Multiplication Rule for Independent events

$$P(A_1 \text{ and } A_2 \text{ and } \dots \text{ and } A_n) = P(A_1)P(A_2) \dots P(A_n)$$

The conditional probability of event A given event B, $P(A|B) = \frac{P(A \cap B)}{P(B)}$

If $A_i; i=1,2,\dots,n$ are mutually exclusive and totally exhaustive n events in a sample space and B is any event of that sample space,

Baye's Theorem, $P(A_i | B) = \frac{P(A_i).P(B|A_i)}{\sum P(A_i).P(B|A_i)}$

$$P(B) = \sum P(A_i).P(B|A_i)$$

Probability Distributions

Expectation, $E(x) = \sum xp(x)$ Variance, $V(x) = E(x^2) - [E(x)]^2$

Normal Distribution $Z = \frac{X - \mu}{\sigma}$

Binomial Distribution $P(x) = {}^nC_x p^x q^{n-x}$; here $q=1-p$, $x=0,1,2,\dots,n$

Poisson Distribution $P(x) = \frac{e^{-\lambda} \lambda^x}{x!}$; here $\lambda = \text{mean}$

Correlation

Karl Pearson's coefficient of correlation, $r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$

Simple linear regression

Equation of regression $y = a + bx$ $b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$ $a = \bar{y} - b\bar{x}$

Coefficient of determination, $R^2 = b^2 \left(\frac{\sum x^2 - n\bar{x}^2}{\sum y^2 - n\bar{y}^2} \right)$ or $R^2 = b^2 \left(\frac{n \sum x^2 - (\sum x)^2}{n \sum y^2 - (\sum y)^2} \right)$ and $R^2 = r^2$